ESWAR VARUN S

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SUMMARY

Software engineer at Accenture building production AI agents and Python backend services on Azure, with the tests, CI/CD and telemetry that keep them reliable. On my own: infrastructure for LLM traffic and tooling for test reliability.

SKILLS

Languages: Python, Java, TypeScript, JavaScript, SQL, C

Backend: FastAPI, Flask, REST APIs, asyncio, Server-Sent Events

AI & LLMs: LangChain, RAG, AI agents, Azure AI Foundry, Azure OpenAI, Gemini API

Cloud & DevOps: Azure, AWS, GCP, Docker, GitHub Actions, Azure DevOps CI/CD, Linux

Data & Testing: MySQL, MongoDB, SQLite, pytest, fault injection, structured logging, telemetry

EXPERIENCE

Advanced Application Engineering Analyst — *Oct 2025 – Present*

Accenture | Chennai, India

Dependency Management Agent *Python, FastAPI, LangChain, Azure DevOps REST API, Microsoft Graph, Azure AI Foundry*

- Built an agent that queries Azure DevOps (WIQL, link-relation traversal) to build a cross-team dependency graph over 100+ concurrent work items in a scaled Agile program, scores at-risk deliverables and routes escalations to the right forum.
- Grounded its answers in program documents with a RAG pipeline over SharePoint/M365 (Microsoft Graph API); content-hash and timestamp change detection reprocesses only what changed between runs.
- Reached >90% accuracy on risk flags, with structured logging, telemetry and alerting to trace and root-cause every mis-scored item.

Portability Agent *Python, FastAPI, LangChain, RAG, Azure AI Foundry, AWS, GCP*

- Built an agent that automates moving applications from AWS and GCP to Azure: a static scanner maps 100+ cloud SDK dependencies in a repository to their Azure equivalents, with RAG over 1,000+ pages of cloud documentation backing each replacement.
- Cut manual migration effort by ∼80% with a >90% migration success rate.
- Owned both agents from design to production: FastAPI services, automated unit, integration and end-to-end tests, Azure DevOps CI/CD, structured logging, telemetry and alerting.

Freelance Software Engineer — *Dec 2024 – Jun 2025*

Outlier (Scale AI) | Remote

- Reviewed LLM-generated solutions to algorithm and software-engineering problems for correctness, time and space complexity and edge cases; wrote reference solutions and grading feedback used in SFT and RLHF training.

PROJECTS

LLM Inference Gateway *Python, FastAPI, asyncio, httpx, pytest* github.com/eswar-varun/llm-inference-gateway

- Built an OpenAI-compatible gateway that routes streaming LLM traffic to Azure OpenAI and any OpenAI-compatible server behind one API; clients adopt it by changing one base URL.
- Designed streaming failure semantics: commit to HTTP 200 only after the upstream's first token so failures get real status codes, and propagate client disconnects upstream, stopping generation within ∼230 ms.
- Classified every upstream failure (timeouts, 429s, 5xx, misconfiguration) into one taxonomy that drives status codes and retryability; 46 tests including fault injection and real-socket integration tests.

FlakeHunter: Flaky-Test Detection *Python, pytest, SQLite, GitHub Actions* github.com/eswar-varun/flakehunter

- Built a tool that ingests JUnit XML from any test runner (pytest, Maven Surefire, Jest), separates flaky tests (pass and fail on the same commit) from broken ones, and ranks them by Wilson-score lower bound.
- Measured on a ground-truth suite of 23 tests (random, timing, async-ordering and order-dependent flakes plus broken tests): 100% precision and 83% recall after 20 runs (mean of 5 trials), re-measured in CI on every push.

CodeXchange: LLM Code Translation *Python, Gemini API, Streamlit, Node.js* github.com/eswar-varun/CodeXchange

- Rebuilt an LLM code translator (12 languages) on the Gemini API after Google retired PaLM; fixed output parsing that leaked language tags into translated code.
- Built an execution-based evaluation: 12 tasks with cross-language traps (negative modulo, banker's rounding, JavaScript's string sort) run the translated code in the target runtime.

EDUCATION & CERTIFICATIONS

B.Tech, Electronics & Computer Engineering, Vellore Institute of Technology, Chennai *2021 – 2025*

CGPA 8.53 | AWS Certified Cloud Practitioner (2024) | Generative AI on Vertex AI, Google & SmartInternz (2024)